



For MOSI, MISO, master out slave in, master in slave out, remember one simple rule and blindly apply the idea that MOSI goes to MISO and MISO goes to MOSI. That may work sometimes, but in the real industry, not everyone follows the same naming convention.

Some vendors use MOSI and MISO. At NXP, we use SIN and SOUT for serial in and serial out. Other vendors use completely different terminology. If you assume instead of verifying, you will make mistakes. So every time I connect a signal to a pin, I go back to the datasheet. Always.

The same applies to unused pins. You cannot simply leave a pin unconnected. The datasheet clearly specifies what to do with unused pins. Some need to be grounded. Some need to be pulled up. Some can be left floating. You cannot do whatever you want. You must follow the recommended condition.

The same care is required with I2C. I2C looks simple. SDA and SCL. But some controllers also provide secondary I2C pins. In certain controllers, these secondary lines are not meant to connect external I2C slaves. If you blindly connect devices to them, communication will fail. You need to read and understand this before connecting anything.

**Watchdog pins in power management ICs.** Many PMICs have an internal watchdog that starts running as soon as the IC powers up. It expects a refresh within a fixed time. Now the obvious question is, if the board is new and the firmware is not ready, how do I refresh the watchdog?

The answer is, I cannot. That is why these ICs provide a pin to disable the watchdog, often described as a debug or development pin. Many engineers ignore it, assuming it is not needed for production. Then the board cannot even boot, and they end up soldering wires and doing ugly hacks just to make it usable. The lesson is simple. Read

every pin description carefully and design based on real usage, not assumptions.

### ***Name your signals properly.***

In large schematics, signals span multiple sheets. If I name a signal UR1TX, it tells me nothing. Instead, I name signals based on source and destination. For example, MCU UR1 to GSM, and GSM to MCU on the return path. Without tracing the net, I already know what the signal does. Good naming makes schematics readable, reviewable, and maintainable.

***Component placement should be physical.*** When I draw a schematic, I place components the way I expect them to sit on the PCB. For an IC power pin, the bulk capacitor can be slightly away, but the bypass capacitor must be very close. Even though schematic placement does not affect electrical behaviour, it strongly influences how the layout engineer thinks. The schematic becomes a visual instruction, not just a connectivity diagram.

***Why do clear schematics matter?*** We learned this the hard way. We had an IC with a battery input, an internal supply pin, and a  $V_{CC}$  pin. It required a bulk capacitor, a 100-nanofarad filter capacitor, and another capacitor on  $V_{CC}$ . Electrically, all three went to ground. The schematic connections were messy but functional, and earlier revisions worked fine.

In a later revision, someone tried to clean up the schematic and removed a single ground connection line. As a result, the filter capacitor and  $V_{CC}$  capacitor were no longer grounded. Electrically, the  $V_{CC}$  pin was now connected through a capacitor instead of being properly bypassed.

The DC-DC converter worked fine up to around 40 to 43 volts, then the output dipped. We were confused because the circuit had worked before. The problem was not the design. It was schematic clarity.

From that point on, we made a

rule. Every component gets its own explicit ground symbol. If an IC has ten capacitors, it gets ten ground connections. No shared nets. No merged lines. Yes, the schematic looks crowded. That is acceptable. What matters is correctness and clarity. Cleaning up schematics just to make them look neat can introduce invisible errors that cost weeks later.

Next, after schematics, we move to the layout stage. The tips are:

***Plan your layout.*** Layout planning starts the moment I begin schematic design. I immediately involve the mechanical team. Connector locations, height restrictions, mounting holes, and enclosure constraints must all be finalised before placement starts.

Heat-generating components are planned carefully. Power devices go where heat can spread or escape. Temperature-sensitive parts like microcontrollers and analogue ICs are placed away from heat sources. Placement is never random.

***Set your rules before routing.*** Before starting PCB design, I talk to fabrication and assembly vendors. Each vendor has different limits on drill sizes, copper clearances, solder mask rules, and silkscreen spacing. If I design without knowing these, I may end up with a board that cannot be manufactured.

The same applies to EMS vendors. Many require a fixed clearance from the board edge where no components are allowed. If this is discovered late, redesign becomes painful or impossible. So I define all rules upfront, board edge clearance, copper rules, and solder mask rules before routing begins.

***Now the artwork begins.*** The schematic was one form of art. PCB layout demands even more attention. I look at how traces flow. Are they straight or unnecessarily zigzagged? How are components oriented? Does the layout look clean and intentional? Visual order often reflects electrical discipline. This is where layout becomes both technical and artistic.